# KeplerLearn

An AI-powered, course-centric learning platform where students get instant, personalized help and teachers see exactly where the class is stuck.

## THE PROBLEM

Static slide decks and office hours don't scale. Students get stuck silently and questions scatter across group chats. Teachers have no visibility into where the class is struggling until the exam is already over.

## WHAT WE BUILT

One app, two roles — Student and Teacher — built around a real course. Students learn slide-by-slide with an AI copilot and get instant, personalized remediation the moment they miss a question. Teachers get a live struggle heatmap and a one-click fix.

## WHAT WE'LL SHOW

**1** **Wrong answer → instant widget**
Student misses an inline quiz question; a personalized interactive widget (Trace Stepper / Slider / Predict & Check) is generated live, grounded in that exact slide.

**2** **Community FAQ → peer content**
A student question is auto-tagged to the right slide; a peer-built widget earns enough upvotes to land in the teacher's review queue.

**3** **Struggle heatmap → one-click fix**
Teacher opens Analytics, sees a color-coded heatmap of where the class is stuck, and generates a supplementary slide in one click.

**4** **Slide deck → structured module**
Teacher runs the Module Setup Wizard: upload decks in, get an ordered lecture sequence with AI-generated practice quizzes and widgets out.

## KEY FEATURES

**AI Copilot**
Cited, slide-aware answers to any question, right inside the lecture.

**Reactive Remediation**
A wrong answer instantly generates a personalized widget, cached class-wide.

**Self-Serve Practice Quizzes**
Students pick any modules or lectures and generate a scoped quiz on demand.

**Community FAQ + Peer Widgets**
Student Q&A and peer-made content, upvoted into the teacher review queue.

**Teacher Struggle Heatmap**
Live, color-coded view of exactly which slides the class is struggling with.

**Module Setup Wizard**
Upload slide decks and get a structured module with AI practice content, ready to publish.

**BUILT ON** Next.js · AWS Fargate · Aurora PostgreSQL (pgvector) · Amazon Bedrock — Claude Sonnet 5 & Haiku 4.5 · Step Functions · Terraform · S3 + CloudFront

**KeplerLearn — where every wrong answer becomes a teaching moment.**